

Indian E-Commerce Sales Analysis using SQL & Excel

Dataset: Public Olist E-Commerce Dataset adapted into an Indian business scenario for portfolio demonstration.

Dataset Size:

Customers: 20,000

Orders: 20,000

Order Items: 20,000

Products: 19,616

Project Overview

This project analyzes e-commerce sales data using SQL and Microsoft Excel to identify revenue trends, top-performing products, category performance, state-wise and regional sales performance, monthly demand trends, and order fulfillment effectiveness through an interactive dashboard.

Business Scenario

Assume you are a Data Analyst at a leading Indian e-commerce company similar to Flipkart or Meesho. Management wants to analyze sales performance, identify top-performing product categories, understand regional revenue trends, and monitor order fulfillment to support business decision-making.

Tools Used

- MySQL Workbench
- Microsoft Excel
- SQL
- Data Visualization

Table Structure Analysis

DESC customers;

DESC orders;

DESC order_items;

DESC products;

SQL File 5*

Limit to 50 rows

```
1 • DESC customers;
2
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
customer_id	text	YES		NULL	
customer_unique_id	text	YES		NULL	
customer_zip_code_prefix	int	YES		NULL	
customer_city	text	YES		NULL	
customer_state	text	YES		NULL	

SQL File 5*

Limit to 50 rows

```
1 • DESC orders;
2
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
order_id	text	YES		NULL	
customer_id	text	YES		NULL	
order_status	text	YES		NULL	
order_purchase_timestamp	text	YES		NULL	
order_approved_at	text	YES		NULL	
order_delivered_carrier_date	text	YES		NULL	
order_delivered_customer_date	text	YES		NULL	
order_estimated_delivery_date	text	YES		NULL	

Result Grid
Form Editor

SQL File 5*

Limit to 50 rows

```
1 • DESC order_items;
2
```

Result Grid | Filter Rows: | Exports: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
order_id	text	YES		NULL	
order_item_id	int	YES		NULL	
product_id	text	YES		NULL	
seller_id	text	YES		NULL	
shipping_limit_date	text	YES		NULL	
price	double	YES		NULL	
freight_value	double	YES		NULL	

Result Grid
Form Editor

SQL File 5* x

Limit to 50 rows

1 • DESC products;

Field	Type	Null	Key	Default	Extra
product_id	text	YES		NULL	
product_category_name	text	YES		NULL	
product_name_lenght	int	YES		NULL	
product_description_lenght	int	YES		NULL	
product_photos_qty	int	YES		NULL	
product_weight_g	int	YES		NULL	
product_length_cm	int	YES		NULL	
product_height_cm	int	YES		NULL	
product_width_cm	int	YES		NULL	

Result 34 x

Read Only

Dataset Validation

Verified the number of records available in each table before performing analysis:

SELECT COUNT(*) FROM customers;

SELECT COUNT(*) FROM orders;

SELECT COUNT(*) FROM order_items;

SELECT COUNT(*) FROM products;

SQL File 5* x

Limit to 50 rows

1 • SELECT COUNT(*) FROM customers;

2 SELECT COUNT(*) FROM orders;

3 SELECT COUNT(*) FROM order_items;

4 SELECT COUNT(*) FROM products;

COUNT(*)
20000

Result Grid

SQL File 5* x

Limit to 50 rows

1 • SELECT COUNT(*) FROM customers;

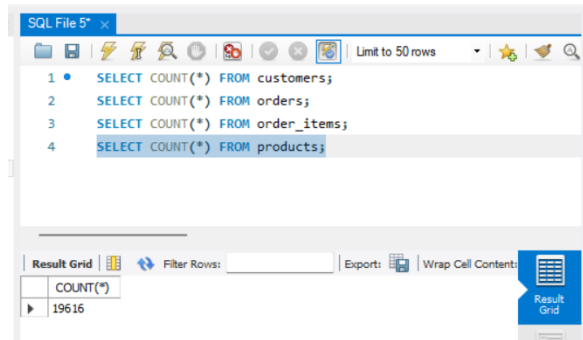
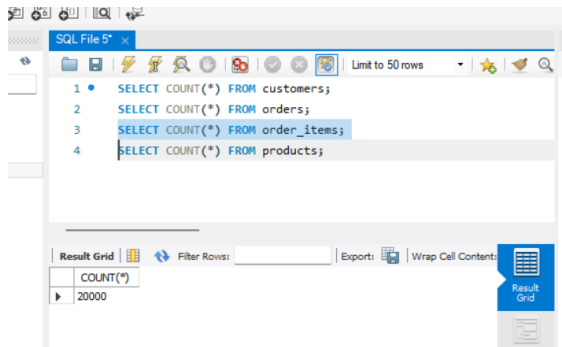
2 SELECT COUNT(*) FROM orders;

3 SELECT COUNT(*) FROM order_items;

4 SELECT COUNT(*) FROM products;

COUNT(*)
20000

Result Grid



Database Schema

The project consists of four primary tables:

- customers – Stores customer location information.
- orders – Contains order details and purchase timestamps.
- order_items – Contains product-level transaction details and pricing.
- products – Contains product category information.

Relationships:

customers → orders → order_items → products

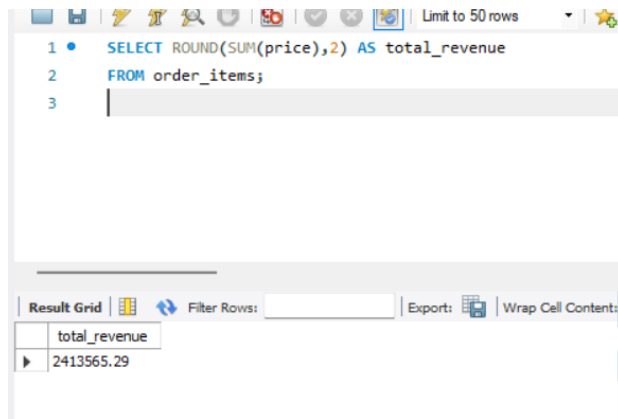
KPI Analysis

Query 1: Total Revenue

Objective: Calculate the total revenue generated from all orders in the dataset.

SQL Query: `SELECT ROUND(SUM(price),2) AS total_revenue FROM order_items;`

Output:



The screenshot shows a SQL query editor with the following query:

```
1 • SELECT ROUND(SUM(price),2) AS total_revenue
2 FROM order_items;
3
```

Below the query editor, the 'Result Grid' is displayed with the following data:

total_revenue
2413565.29

Business Insight: The total revenue generated from the analyzed dataset was ₹241,356.29. This KPI provides an overall view of business performance and serves as the foundation for further sales analysis.

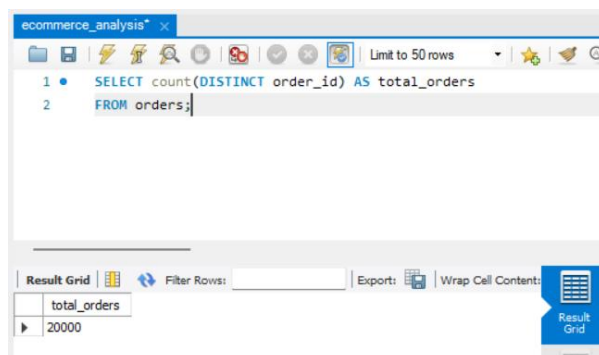
Query 2: Total Orders

Objective

Determine the total number of orders available in the dataset.

SQL Query: SELECT count(DISTINCT order_id) AS total_orders FROM orders;

Output:



The screenshot shows a SQL query editor with the following query:

```
1 • SELECT count(DISTINCT order_id) AS total_orders
2 FROM orders;
```

Below the query editor, the 'Result Grid' is displayed with the following data:

total_orders
20000

Business Insight: A total of 20,000 orders were analyzed. This metric helps measure transaction volume and provides context for revenue and customer activity analysis.

Query 3: Average Order Value

Objective: Calculate the average revenue generated per order.

SQL Query: SELECT ROUND(SUM(price)/COUNT(DISTINCT order_id),2) AS avg_order_value FROM order_items;

Output:

The screenshot shows a SQL query in a tool named 'ecommerce_analysis'. The query is: `SELECT ROUND(SUM(price)/COUNT(DISTINCT order_id),2) AS avg_order_value FROM order_items;`. The result grid below shows a single row with the value 137.55.

avg_order_value
137.55

Business Insight: The average order value was ₹137.55, indicating the average amount spent per transaction. This metric helps businesses understand customer purchasing behavior and evaluate sales performance.

Revenue Analysis

Query 4: Top 10 Products by Revenue

Objective: Identify the highest revenue-generating products to understand which products contribute most to overall sales.

SQL Query: `SELECT product_id, ROUND(SUM(price),2) AS revenue FROM order_items GROUP BY product_id ORDER BY SUM(price) DESC LIMIT 10;`

Output:

The screenshot shows a SQL query in a tool named 'ecommerce_analysis'. The query is: `SELECT product_id, ROUND(SUM(price),2) AS revenue FROM order_items GROUP BY product_id ORDER BY SUM(price) DESC LIMIT 10;`. The result grid below shows the top 10 products by revenue.

product_id	revenue
5769ef0a239114ac3a854af0df129e4	13440
bb50f2e236e5ee80100680137654686c	11805
6cdd53843498f92890544667809f1595	11631.3
d6160fb7873f184099d9bc95e30376af	8399.97
588531f8ec37e7d5ff5b7b22ea0488f8	8148
d1c427060a0f73f6b889a5c7651f2ac4	8009.73
99a4788cb24856965c36a24e33966058	7287.2
489ae2aa008f021502940f251d4cce7f	6735
aca2eb7800ea1a7b8eb44e68314663af	6428.7
3dd2a17168ec895c781a9191c1e95ad7	6295.8

Business Insight: The top-performing product generated approximately ₹13,440 in revenue, indicating that a small group of products contributes significantly to overall business revenue.

Query 5: Top Categories by Revenue

Objective: Identify product categories contributing the highest revenue.

Tables Used:

- order_items
- products

SQL Concepts Used: INNER JOIN, GROUP BY, SUM, ORDER BY

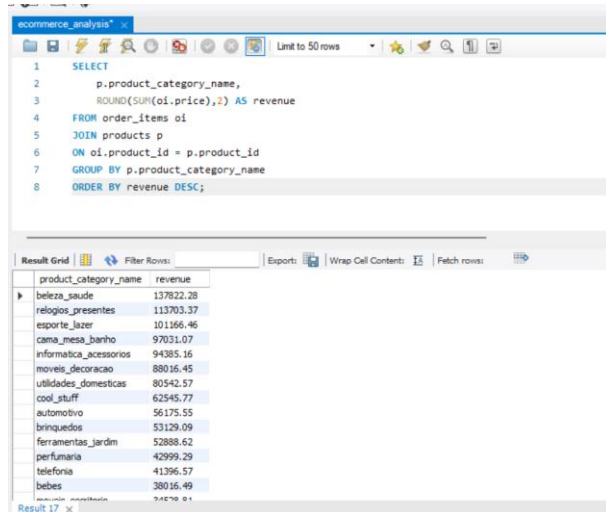
SQL Query: SELECT

p.product_category_name,

ROUND(SUM(oi.price),2) AS revenue

FROM order_items oi JOIN products p ON oi.product_id = p.product_id GROUP BY
p.product_category_name ORDER BY revenue DESC;

Output:



The screenshot shows a SQL query editor window titled 'ecommerce_analysis'. The query is as follows:

```
1 SELECT
2   p.product_category_name,
3   ROUND(SUM(oi.price),2) AS revenue
4 FROM order_items oi
5 JOIN products p
6 ON oi.product_id = p.product_id
7 GROUP BY p.product_category_name
8 ORDER BY revenue DESC;
```

Below the query, the results are displayed in a table with two columns: 'product_category_name' and 'revenue'. The results are ordered by revenue in descending order.

product_category_name	revenue
beleza_saude	137622.28
religios_presentes	113703.37
esporte_lazer	101166.46
cama_mesa_banho	97031.07
informatica_acessorios	94385.16
moveis_decoracao	88016.45
utilidades_domesticas	80542.57
cool_stuff	62545.77
automotivo	56175.55
brinquedos	53129.09
ferramentas_jardim	52888.62
perfumaria	42999.29
telefonia	41396.57
bebidas	38016.49

Business Insight: Product category analysis revealed that a few categories contribute a significant portion of total revenue. These categories should be prioritized for inventory management, promotions, and marketing campaigns.

Trend Analysis

Query 6: Monthly Order Trend

Objective: Analyze month-wise order volume to identify growth patterns and seasonal trends.

SQL Concepts Used: STR_TO_DATE, DATE_FORMAT, GROUP BY, COUNT, ORDER BY

SQL Query: SELECT DATE_FORMAT(

STR_TO_DATE(order_purchase_timestamp,'%d-%m-%Y %H:%i'),

'%Y-%m'

) AS month, COUNT(*) AS total_orders FROM orders GROUP BY month ORDER BY month;

Output:

The screenshot shows a SQL query in a window titled 'e-commerce_analysis'. The query is as follows:

```

1 SELECT
2   DATE_FORMAT(
3     STR_TO_DATE(order_purchase_timestamp, '%d-%m-%Y %H:%i:%s'),
4     '%Y-%m'
5   ) AS month,
6
7   COUNT(*) AS total_orders
8
9 FROM orders
10

```

Below the query is a 'Result Grid' with the following data:

month	total_orders
2017-08	904
2017-09	943
2017-10	889
2017-11	1569
2017-12	1136
2018-01	1451
2018-02	1385
2018-03	1443
2018-04	1360
2018-05	1391
2018-06	1215
2018-07	1266
2018-08	1336
2018-09	4

Business Insight: Monthly order analysis showed fluctuations in customer demand over time, helping identify potential seasonal trends and business growth patterns.

Regional Analysis

Query 7: State-wise Revenue Analysis

Objective: Identify the regions contributing the highest sales revenue.

SQL Concepts Used: Multiple JOINS, GROUP BY, SUM, ORDER BY

SQL Query:

```

SELECT

    c.customer_state,

    ROUND(SUM(oi.price),2) AS revenue

FROM customers c JOIN orders o

    ON c.customer_id = o.customer_id

JOIN order_items oi

    ON o.order_id = oi.order_id

GROUP BY c.customer_state ORDER BY revenue DESC;

```

Output:

The screenshot shows a SQL query in a window titled 'ecommerce_analysis'. The query is as follows:

```

1 SELECT
2   c.customer_state,
3   ROUND(SUM(oi.price),2) AS revenue
4 FROM customers c
5 JOIN orders o
6   ON c.customer_id = o.customer_id
7 JOIN order_items oi
8   ON o.order_id = oi.order_id
9 GROUP BY c.customer_state
10 ORDER BY revenue DESC;

```

Below the query, the 'Result Grid' shows the top 21 states by revenue:

customer_state	revenue
SP	39453.58
RJ	28645.03
MG	11754.82
SC	8786.35
CE	4233.83
RS	3633.82
PR	3423.25
BA	2859.1
TO	2779.93
MA	1631.16
SE	1543.9
GO	1390.36
MT	1301.57
MS	1027.99
PA	732.6
AL	682.6
DF	678.6
AM	678.6
RN	678.6
RO	678.6
PE	678.6
AC	678.6

Business Insight: Revenue was concentrated in a few regions, with certain states generating significantly higher sales than others. This insight can help businesses prioritize marketing efforts and resource allocation in high-performing regions

Query 8: Regional Revenue Analysis

Objective: Group Indian states into business regions and identify the highest revenue-generating regions.

Business Question Answered: Which business regions contribute the highest revenue?

SQL Concepts Used: CASE, INNER JOIN, GROUP BY, SUM, ORDER BY

SQL Query:

SELECT

CASE

WHEN c.customer_state IN ('SP','GO','CE')

THEN 'West'

WHEN c.customer_state IN ('RJ','SC','TO','MA','PB')

THEN 'South'

WHEN c.customer_state IN ('MG','RS','SE','DF','AM','RN','RO','PE')

THEN 'North'

WHEN c.customer_state IN ('PR','MS','BA','PA','AL')

THEN 'East'

WHEN c.customer_state IN ('MT','ES')

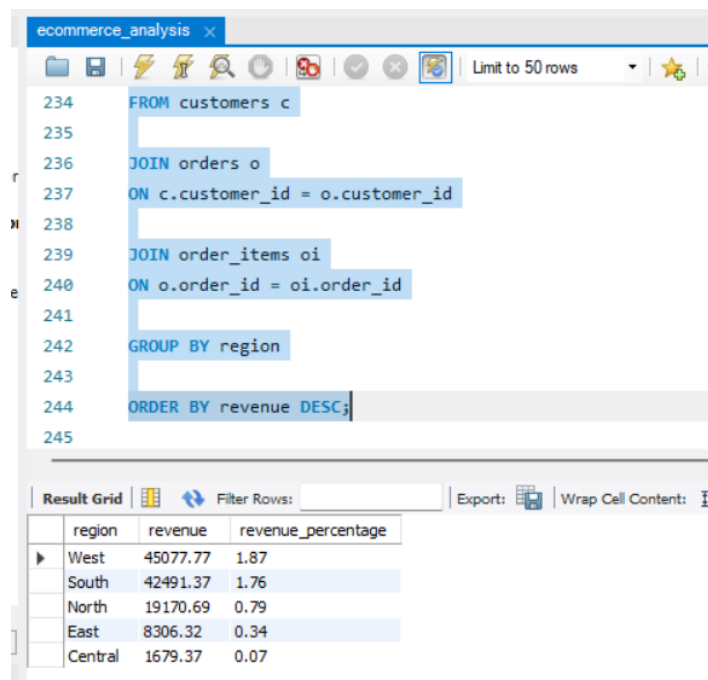
THEN 'Central'

```

END AS region,
ROUND(SUM(oi.price),2) AS revenue,
ROUND(
    SUM(oi.price) * 100 /
    (SELECT SUM(price) FROM order_items),
    2
) AS revenue_percentage
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
JOIN order_items oi
ON o.order_id = oi.order_id
GROUP BY region
ORDER BY revenue DESC;

```

Output:



The screenshot shows a SQL IDE window titled 'ecommerce_analysis'. The query is highlighted in blue, and the results are shown in a table below. The table has four columns: region, revenue, and revenue_percentage. The results are ordered by revenue in descending order.

region	revenue	revenue_percentage
West	45077.77	1.87
South	42491.37	1.76
North	19170.69	0.79
East	8306.32	0.34
Central	1679.37	0.07

Business Insight: West and South regions generated the highest revenue among all business regions. This analysis helps identify high-performing markets and supports regional marketing, inventory planning, and business expansion strategies.

Query 9: Order Status Distribution

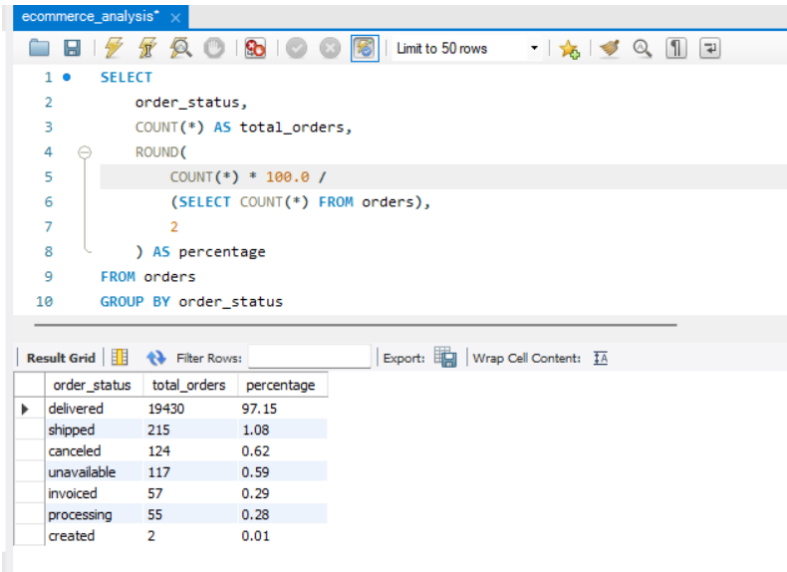
Objective: Analyze order fulfillment performance and identify the distribution of different order statuses.

SQL Concepts: GROUP BY, COUNT, Subquery, Percentage Calculation, ORDER BY

SQL Query:

```
SELECT
    order_status,
    COUNT(*) AS total_orders,
    ROUND(
        COUNT(*) * 100.0 /
        (SELECT COUNT(*) FROM orders),
        2
    ) AS percentage
FROM orders GROUP BY order_status ORDER BY total_orders DESC;
```

Output:



The screenshot shows a SQL query editor window titled 'ecommerce_analysis'. The query is as follows:

```
1 SELECT
2   order_status,
3   COUNT(*) AS total_orders,
4   ROUND(
5     COUNT(*) * 100.0 /
6     (SELECT COUNT(*) FROM orders),
7     2
8   ) AS percentage
9 FROM orders
10 GROUP BY order_status
```

Below the query editor, the 'Result Grid' shows the output of the query. The table has three columns: 'order_status', 'total_orders', and 'percentage'. The data is sorted by 'total_orders' in descending order.

order_status	total_orders	percentage
delivered	19430	97.15
shipped	215	1.08
canceled	124	0.62
unavailable	117	0.59
invoiced	57	0.29
processing	55	0.28
created	2	0.01

Business Insight: The analysis revealed that the majority of orders were successfully delivered, while only a small percentage were cancelled, shipped, or in other processing stages. This indicates strong operational efficiency and effective order fulfillment performance.

Query 10: State-wise Revenue Contribution Analysis

Objective: Measure each region's contribution to overall revenue and identify top-performing locations

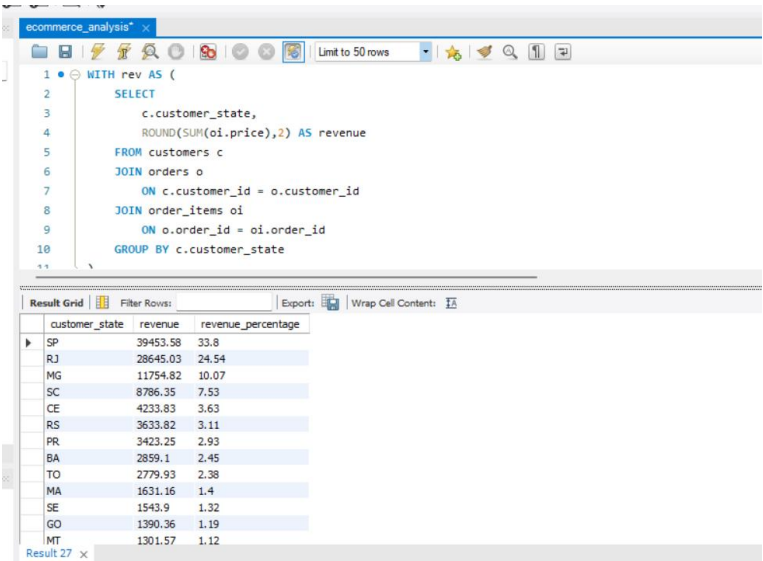
Business Question Answered: The top revenue-generating regions contributed a substantial share of total revenue, indicating that sales performance is concentrated in specific geographical areas.

SQL Concepts Used: CTE (WITH), INNER JOIN ,GROUP BY, SUM, Subquery, Percentage Calculation, ORDER BY

SQL Query:

```
WITH rev AS (  
    SELECT  
        c.customer_state,  
        ROUND(SUM(oi.price),2) AS revenue  
    FROM customers c  
    JOIN orders o  
        ON c.customer_id = o.customer_id  
    JOIN order_items oi  
        ON o.order_id = oi.order_id  
    GROUP BY c.customer_state  
)  
SELECT  
    customer_state,  
    revenue,  
    ROUND(  
        revenue * 100 /  
        (SELECT SUM(revenue) FROM rev),  
        2  
    ) AS revenue_percentage FROM rev ORDER BY revenue DESC;
```

Output:



```
1 WITH rev AS (  
2   SELECT  
3     c.customer_state,  
4     ROUND(SUM(oi.price),2) AS revenue  
5   FROM customers c  
6   JOIN orders o  
7     ON c.customer_id = o.customer_id  
8   JOIN order_items oi  
9     ON o.order_id = oi.order_id  
10  GROUP BY c.customer_state
```

customer_state	revenue	revenue_percentage
SP	39453.58	33.8
RJ	28645.03	24.54
MG	11754.82	10.07
SC	8786.35	7.53
CE	4233.83	3.63
RS	3633.82	3.11
PR	3423.25	2.93
BA	2859.1	2.45
TO	2779.93	2.38
MA	1631.16	1.4
SE	1543.9	1.32
GO	1390.36	1.19
MT	1301.57	1.12

Business Insight: The majority of orders were successfully delivered, indicating strong order fulfillment performance and operational efficiency.

Dashboard Development

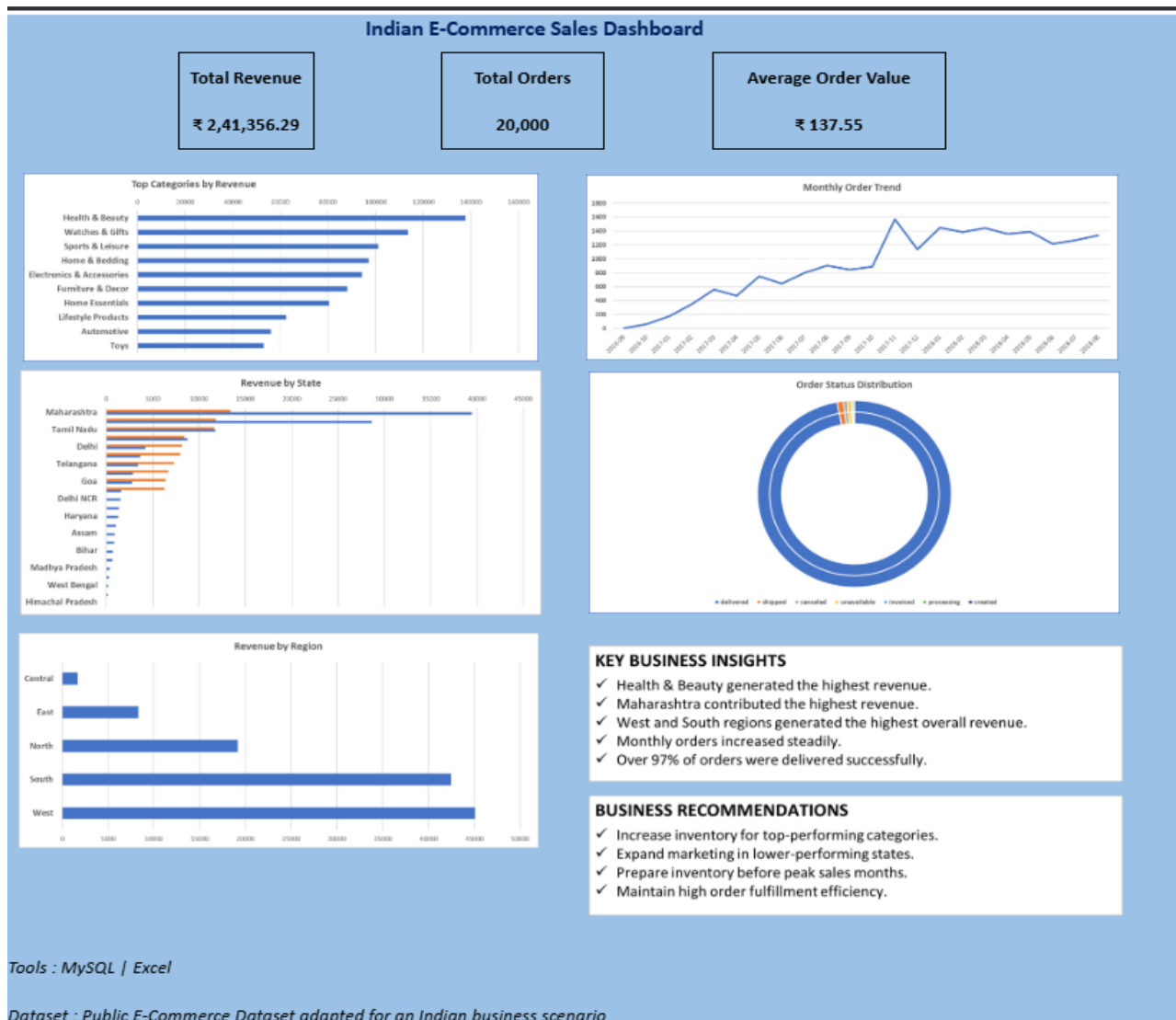
Dashboard Overview

An interactive Excel dashboard was developed using the SQL analysis outputs to visualize key business metrics and performance indicators.

Dashboard Components

- Total Revenue
- Total Orders
- Average Order Value
- Top Categories by Revenue
- Monthly Order Trend
- Regional Revenue Analysis
- Order Status Distribution

Dashboard Screenshot



Tools : MySQL | Excel

Dataset : Public E-Commerce Dataset adapted for an Indian business scenario

Key Findings

1. Total revenue generated from the analyzed dataset was ₹241,356.29 across 20,000 orders.
2. Health & Beauty generated the highest revenue among all product categories.
3. Maharashtra contributed the highest revenue among all analyzed states.
4. West and South regions generated the highest overall revenue.
5. Monthly order trends indicated increasing customer demand over time.
6. More than 97% of orders were successfully delivered, reflecting strong operational performance.
7. Product revenue was concentrated among a relatively small number of categories and products.

8. SQL joins, aggregations, CTEs, CASE statements, subqueries, and date functions were used to generate business insights and build the dashboard.

Conclusion

This project analyzed e-commerce sales data using SQL to evaluate revenue performance, product effectiveness, category contribution, regional sales distribution, monthly demand trends, and order fulfillment metrics. Through the use of SQL techniques such as joins, aggregations, CTEs, subqueries, and date functions, meaningful business insights were extracted from transactional data.

The analysis identified key revenue-generating products and categories, highlighted regional sales patterns, and assessed operational performance through order status evaluation. These insights can support data-driven decision-making in areas such as inventory planning, marketing strategy, regional expansion, and customer experience improvement.

Overall, this project demonstrates practical Data Analyst skills in data exploration, SQL-based business analysis, KPI development, and insight generation using a public e-commerce dataset adapted into an Indian business scenario.

This project demonstrates an end-to-end data analytics workflow, beginning with data exploration in MySQL, followed by SQL-based business analysis, Excel dashboard development, and business insight generation. The project showcases practical skills in SQL querying, KPI reporting, dashboard creation, and business storytelling using a public e-commerce dataset adapted to an Indian business scenario.

Future Enhancements

- Build an interactive Power BI dashboard.
- Perform Customer Segmentation.
- Implement RFM Analysis.
- Develop Sales Forecasting models.
- Analyze Customer Retention and Cohort Trends.
- Automate reporting using SQL and Power BI.

Prepared By:

Abhineeth K

Tools:

MySQL Workbench | Microsoft Excel